# Conditional zero order at an actual projective branch

Formal proof packet

27 October 2026

This packet records a conditional local-geometry proof, not a target experiment or cost comparison. The editable diagram, gate CSV, Lean source, and compiler log are beside this PDF.

## Checked statement

Let $a$ be a rational simple root of the quintic and $R$ the structure-sheaf stalk of the actual weighted-Proj curve at that branch. The previously checked local geometry makes $R$ a DVR, with $\mathrm{ord}(y) = 1$ and $y^2 = (x - a)q$ for a unit $q$. The new theorem proves that if $N = (x - a)t$ and $B$ is a unit in $R$, then

$$\mathrm{ord}(N - By) = 1.$$

The proof transports the generic branch zero-order lemma into this actual projective stalk. The underlying DVR criterion is discussed in the Stacks Project, Lemma 10.119.7.

A second checked theorem composes the separate norm-factor transport with this result. If a polynomial norm identity holds at a rational support root and $B$ is a unit in this stalk, then the evaluated polynomial numerator satisfies $\mathrm{ord}(n(x) - By) = 1$. A checked corollary replaces the unit premise with the source-style scalar guard $b \neq 0$: its image $B$ is a unit. The norm polynomial and its link to the compiled runtime numerator remain explicit inputs.

Checked actual stalk: $R$ is a DVR and $\mathrm{ord}(y) = 1$

Explicit input: $N = (x - a)t$ (or the checked norm route), $b \neq 0$

$\Downarrow$ checked local valuation algebra

$\mathrm{ord}(N - By) = 1$

## Verification and remaining gates

Direct Lean 4.34.1 compilation with Mathlib `d13f23b` exited zero for both sources. All four theorem axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Local source SHA-256:

`49005acc564b41341b37296d42de90c48e56ac4f4a4ec4860d7278fc8bc00cca.`

Composition source SHA-256: `8c136b70da26dc928a3ea1b1646dd4ff40d28240b52ea1b7ae1c561301ff35f5.`

The compiled Rust law has not been shown to equal the mapped polynomial or satisfy the unit hypothesis at every branch. Nonbranch multiplicities, the global principal divisor, the $\mathrm{Pic}^0$ class equation, and universal runtime semantics remain open. No frozen 1,260-target row or matched-rho ratio changes.